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PROJECT DOCUMENTATION

Project	***Hephaestus***
Version	1.0.0
Date	May 12, 2026
Author	Mario Suarez Ayala
Repository	https://github.com/MarioSuarezAyala1/3D-Design-Library-210-Models-for-Additive-Manufacturing
Type	3D Design / Additive Manufacturing / Render
Status	Stable

📋 Project Overview

Hephaestus is a large-scale experimental project focused on mass production of optimized 3D printable models for online distribution and monetization through digital marketplaces such as Cults3D.

The objective of the project was to validate whether artificial intelligence workflows combined with CAD and rendering pipelines could accelerate the development of commercially viable 3D printable assets. The workflow successfully enabled the creation of over 100 printable models while maintaining visual quality, printability and scalability.

The process involved analyzing real-world objects using AI-assisted reference extraction, generating profile and side-view blueprints, rebuilding the models in Blender, refining geometry with additional AI-assisted detailing tools, and producing professional renders for marketplace publication.

The project currently achieved:

- **102 published models**
- **92 total sales**
- **€380.61 total earnings**
- **More than 1400 downloads**
- Commercial deployment through Cults3D

The project is intended for:

- Makers and hobbyists
- 3D printing enthusiasts
- Digital product marketplaces
- Rapid prototyping experimentation
- AI-assisted design workflow research

⚙️ Technical Specifications

Fill in the table with the key system parameters. Add or remove rows according to the project type.

Parameter	Value / Description	Unit
Main Software Platform	Blender / AI-assisted workflow	—
Operating System	Windows 11	—
Primary Modeling Software	Blender	—
AI Reference Generation	Google Gemini	—
AI Detail Refinement	Meshy AI / AI-assisted mesh refinement	—
Render Engine	Blender Cycles	—
Communication Protocol	Git / Cloud Sync	—
Average Model Polycount	Optimized low-mid poly printable meshes	—
Output Formats	STL / OBJ / FBX / PNG	—
Marketplace Platform	Cults3D	—
Number of Models	102 Published Models	—
Total Downloads	1400+	—
Total Sales	92	—
Total Revenue	€380.61	EUR
Render Resolution	1920 × 1080	px
Design Pipeline	AI → Blender → AI Refinement → Blender Render	—

🏗️ System Architecture

Mechanical Subsystem

The mechanical subsystem consisted of the development of optimized printable geometries intended for FDM additive manufacturing. Models were designed to minimize unsupported overhangs while maintaining structural integrity and visual appeal.

The workflow emphasized:

- Printable topology
- Reduced support usage
- Modular geometry cleanup
- Mesh optimization
- Dimensional consistency

Manufacturing process:

- FDM 3D printing

- STL export optimization
- Cura slicing validation
- Test print verification

Electronic Subsystem

This project did not require embedded electronics directly, but the hardware ecosystem involved:

- Desktop workstation for rendering and modeling
- GPU acceleration for Blender Cycles rendering
- Cloud-based AI processing tools

Software / Firmware Subsystem

The software architecture followed a multi-stage pipeline:

1. Real-world object reference collection
2. AI-assisted profile blueprint generation using Google Gemini
3. Blender-based modeling reconstruction
4. AI-assisted geometry enhancement using Meshy AI
5. Manual cleanup and detailing in Blender
6. Material assignment and scene composition
7. Multi-angle rendering export
8. Marketplace optimization and upload

Implemented techniques:

- Mesh retopology
- Boolean operations
- Non-destructive modifiers
- Render compositing
- Asset optimization

Render / Visualization

Render / Visualization

Rendering was performed using Blender Cycles with studio-style lighting setups.

Typical render pipeline:

- HDRI environment lighting
- Metallic and plastic PBR materials
- Depth of field configuration
- Multiple camera angle exports
- Shadow catcher integration

Each model included approximately:

- 5 render angles
- Marketplace thumbnail optimization
- High-contrast studio composition

🔬 Process & Methodology

The project originated from the need to experimentally validate whether AI-assisted workflows could dramatically increase productivity in commercial 3D asset generation.

Functional Requirements

- Produce printable 3D models rapidly
- Maintain commercial-quality renders
- Optimize models for marketplace publication
- Generate scalable production workflow

Non-Functional Requirements

- Low production cost
- Fast iteration cycles
- Minimal hardware investment
- Consistent render quality
- File compatibility with common slicers

Planning Tools

- Blender
- Google Gemini
- GitHub
- Cults3D
- Cloud AI tools

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Conceptual Design

The initial concept explored combining AI-generated references with manual 3D modeling workflows.

Instead of fully AI-generated meshes, the selected approach used AI as an accelerated concept and blueprint generator. Google Gemini was used to analyze real-world objects and generate side and profile references that simplified the reconstruction process.

This method was selected because:

- It reduced modeling time significantly
- It preserved artistic control
- It improved topology quality compared to raw AI meshes
- It allowed scalable production

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Detailed Design

Detailed design was performed primarily in Blender.

Main design decisions:

- Use Blender due to flexibility and speed
- Maintain printable wall thicknesses
- Optimize polygon counts
- Export standardized STL files
- Use AI refinement only after primary geometry reconstruction

Additional refinement tools such as Meshy AI were used to:

- Improve small surface details
- Enhance organic geometry
- Smooth irregular topology
- Accelerate secondary detailing

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Prototyping & Fabrication

Prototype workflow:

1. Create reference images
2. Generate blueprint-style side and profile references
3. Model geometry in Blender
4. Refine details with AI-assisted tools
5. Return to Blender for final cleanup
6. Export STL files
7. Validate in slicer software
8. Produce renders
9. Upload to marketplace

Materials and tools used:

- PLA validation prints
- Cura slicer
- Blender Cycles rendering
- AI cloud processing

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Programming & Integration

Although the project did not include embedded firmware, integration focused on workflow automation and interoperability between tools.

Integrated workflow:

- AI-generated references
- Blender geometry pipeline
- Mesh refinement tools
- Render automation
- Marketplace asset packaging

Configured formats:

- STL

- OBJ
- PNG
- FBX

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Testing & Validation

Validation methods included:

- STL integrity testing
- Non-manifold geometry inspection
- Cura slicing verification
- Support requirement analysis
- Printability validation

Performance metrics:

- 102 commercially released models
- 92 successful sales
- More than 1400 downloads
- Positive marketplace engagement

Challenges encountered:

- Geometry cleanup after AI refinement
- Maintaining printable topology
- Balancing detail and polygon optimization

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Render & Documentation

Rendering software:

- Blender Cycles

Scene configuration:

- HDRI studio lighting
- Product-style camera composition
- PBR materials
- Neutral backgrounds

Documentation structure:

- Marketplace-ready descriptions
- Render galleries
- STL packaging
- GitHub repository organization

Tools & Equipment Required

Category	Tool / Equipment	Notes
Mechanical	Blender / Cura / 3D Printer	Blender 4.x
Electronic	Desktop Workstation	GPU accelerated
Software	Google Gemini / GitHub	Cloud AI workflow
Render	Blender Cycles	Physically based rendering
Other	Cults3D Marketplace	Commercial distribution

Key Decisions & Lessons Learned

- ### Decision 1

- Use AI only for references and refinement rather than complete generation to maintain control over topology and printability.

- ### Decision 2

- Standardize render angles and marketplace presentation to improve commercial consistency.

- ### Challenge Encountered

- AI-refined meshes frequently produced topology artifacts and non-manifold geometry. This was solved through manual cleanup and retopology in Blender.

- ### What Would Be Done Differently

- Future versions could integrate procedural automation and batch rendering pipelines to further increase scalability.

🔄 Model Replacement Guide

Step-by-step instructions for replacing, updating or adapting the models (3D parts, electronic modules, firmware, or renders) used in this project.

> ⚠️ **Before you start:** Back up the original files and note the current version before making any replacements.

3D / Mechanical Model Replacement

#	Action	Tool	Notes
1	Locate original Blender file	Blender	File path: /models/[model_name].blend
2	Verify printable geometry	Blender analysis tools	Check wall thickness
3	Import or redesign model	Blender	Maintain origin alignment
4	Export STL	Blender export	Binary STL preferred
5	Slice and validate	Cura	Check supports
6	Update render scene	Blender Cycles	Re-apply lighting
7	Upload updated assets	Cults3D / GitHub	Update thumbnails

Electronic Module Replacement

Not applicable for this project.

Firmware / Software Module Replacement

#	Action	Tool	Notes
1	Identify AI tool to replace	Workflow documentation	Note compatibility
2	Validate exported formats	Blender	STL/OBJ compatibility
3	Update workflow pipeline	Documentation	Maintain naming structure
4	Test rendering output	Blender Cycles	Compare quality
5	Validate marketplace assets	Cults3D	Verify previews
6	Commit repository updates	GitHub	Tag new version

> ✅ **After any replacement:** Update the Version field in the metadata table at the top of this document and commit all modified files to the repository.

🔩 Bill of Materials (BOM)

Component	Specification	Qty	Reference
Blender	3D Modeling Software	1	Blender Official Website
Google Gemini	AI Reference Generation	2	Google Gemini
Meshy AI	AI Mesh Refinement	1	Meshy AI
Cura	Slicer Software		UltiMaker Cura
Cults3D	Marketplace Platform		Cults3D

🚀 Installation & Usage

1. Clone the repository:
   git clone https://github.com/mario-suarez-ayala/[repo].git
2. Install dependencies (if applicable):
   [installation command]
3. Flash the firmware / run the software:
   [command or tool used]

📊 Results & Testing

Project achievements:

- 102 uploaded commercial models
- 92 total sales
- €380.61 total revenue
- More than 1400 downloads
- Stable AI-assisted workflow validated

The experiment demonstrated that AI-assisted design pipelines can significantly accelerate 3D asset production while preserving sufficient quality for commercial marketplaces.

The project also validated:

- AI-assisted concept extraction
- Hybrid manual/AI modeling workflows
- Scalable digital manufacturing pipelines
- Marketplace viability of mass-produced printable assets

🤝 Contributing

Contributions are welcome. Please open an Issue before submitting a Pull Request. Maintain naming conventions, render consistency and printable topology standards.

📄 License

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📬 Contact

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